

Problem

When port 1 of a two-port circuit is short-circuited, $I_1 = 4I_2$ and $V_2 = 0.25I_2$. Which of the following is true?

- (a) $y_{11} = 4$
- (b) $y_{12} = 16$
- (c) $y_{21} = 16$
- (d) $y_{22} = 0.25$

Step-by-step solution

Step 1 of 1

$$I_1 = 4I_2$$

$$V_2 = 0.25 I_2$$

$$Y_{11} = \left. \frac{I_1}{V_1} \right|_{V_2=0} = 4 \text{ 'Not True'}$$

$$Y_{12} = \left. \frac{I_1}{V_2} \right|_{V_1=0} = \frac{4I_2}{0.25I_2} = 16 \text{ True}$$

$$Y_{21} = \left. \frac{I_2}{V_1} \right|_{V_2=0} = 16 \text{ Not True}$$

$$Y_{22} = \left. \frac{V_2}{I_1} \right|_{I_1=0} = 0.25 \text{ Not True}$$